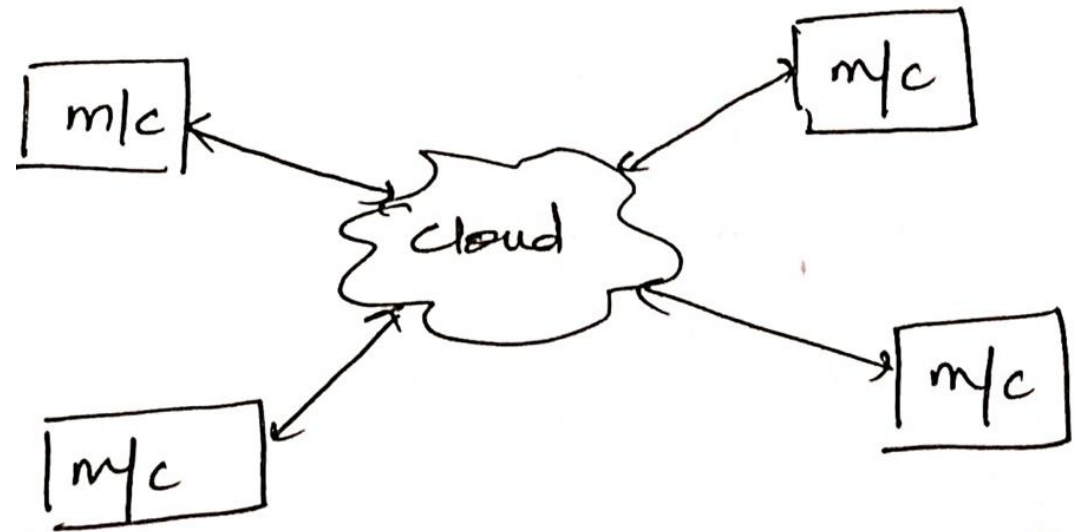
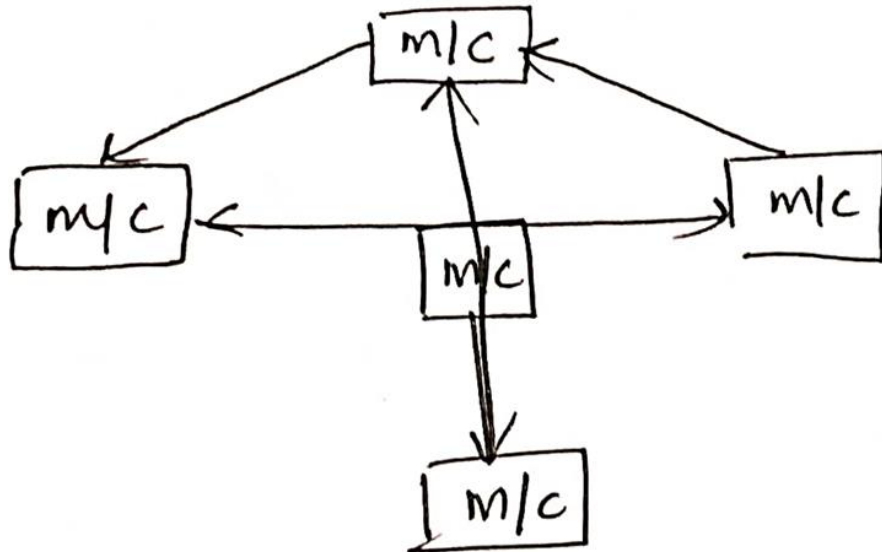


Module 2:

SENSOR NETWORKS

Machine-to-Machine Communication

- M2M refers to direct communication between devices or machines using wired or wireless communication channels. It typically involves sensors, meters, or other devices exchanging data without human intervention. M2M often operates within closed systems or specific applications.
- IoT extends beyond M2M by incorporating connectivity not just between machines, but also between machines and applications, services, or people. IoT often involves a broader ecosystem of interconnected devices, networks, and platforms enabling diverse applications and services.



M2M vs IoT Example

Assume a coffee machine on each floor of a five-floor building:

CASE-I: No Connectivity

CASE-II: Basic Connectivity (M2M) – Machine has basic sensors & it can send notifications phone, e-mail, SMS etc. whenever coffee beans, chocolate powder, sugar, milk powder etc. falls below a certain level.

CASE-III: Advanced Connectivity (IoT) – machines connected to office systems and database.

CASE-IV: Intelligent IoT - It can even know about the inventory for stocking the material in time.

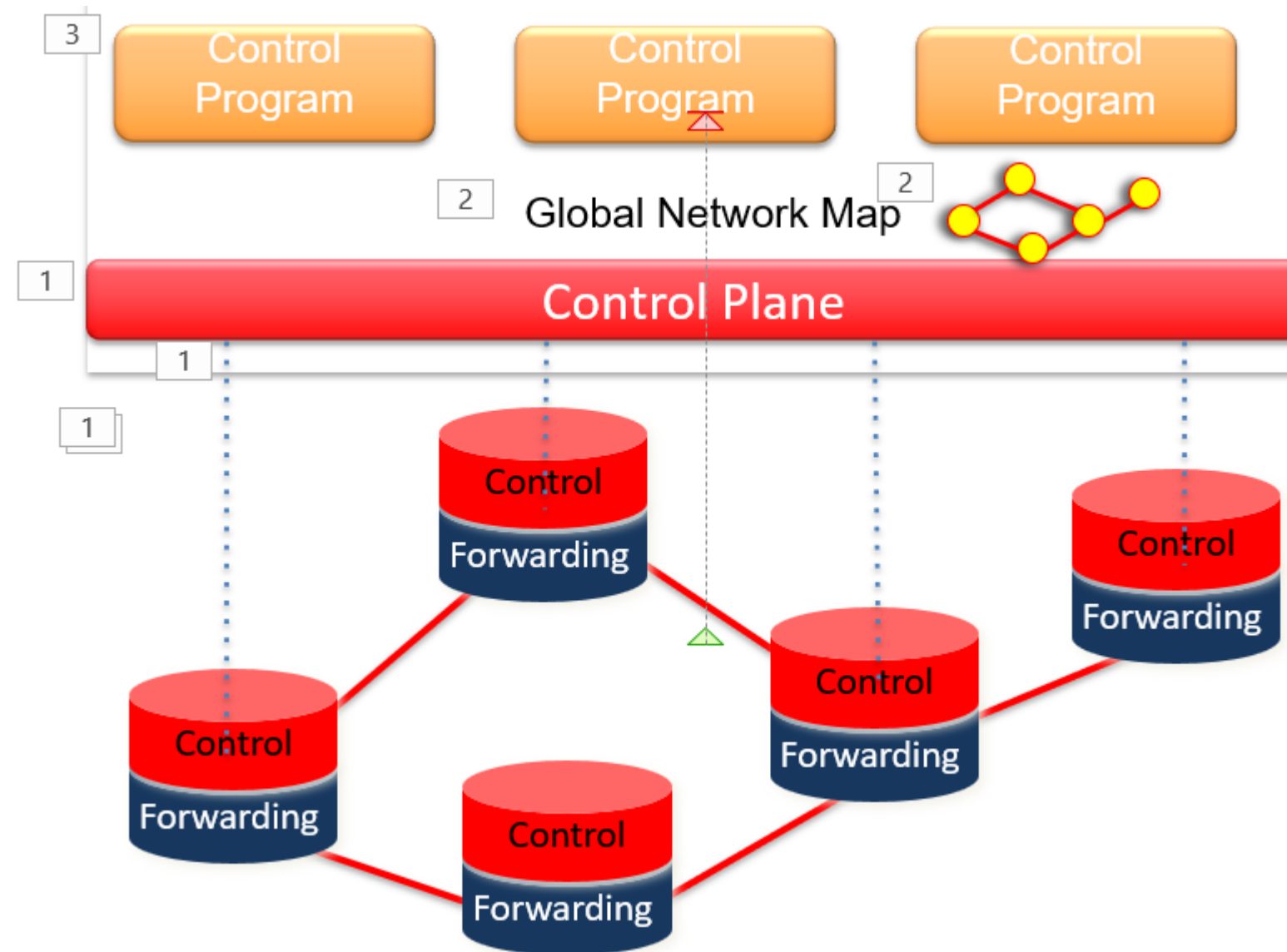
M2M vs IoT

Basis of Difference	Internet of Things	Machine to Machine
Intelligence	Devices have objects that are responsible for decision making	Some degree of intelligence is observed in this.
Connection type used	The connection is via Network and using various communication types.	The connection is a point to point
Communication protocol used	Internet protocols are used such as HTTP, FTP and Telnet.	Traditional protocols and communication technology techniques are used
Data Sharing	Data is shared between other applications that are used to improve the end-user experience.	Data is shared with only the communicating parties.
Internet	Internet connection is required for communication	Devices are not dependent on the Internet.
Type of Communication	It supports cloud communication	It supports point-to-point communication.
Computer System	Involves the usage of both Hardware and Software.	Mostly hardware-based technology
Scope	A large number of devices yet scope is large.	Limited Scope for devices.
Business Type used	Business 2 Business(B2B) and Business 2 Consumer(B2C)	Business 2 Business (B2B)
Open API support	Supports Open API integrations.	There is no support for Open APIs
It requires	Generic commodity devices.	Specialized device solutions.
Centric	Information and service centric	Communication and device centric.
Approach used	Horizontal enabler approach	Vertical system solution approach .
Components	Devices/sensors, connectivity, data processing, user interface	Device, area networks, gateway, Application server.
Examples	Smart wearables, Big Data and Cloud, etc.	Sensors, Data and Information, etc.

Software Defined Networks (SDN)

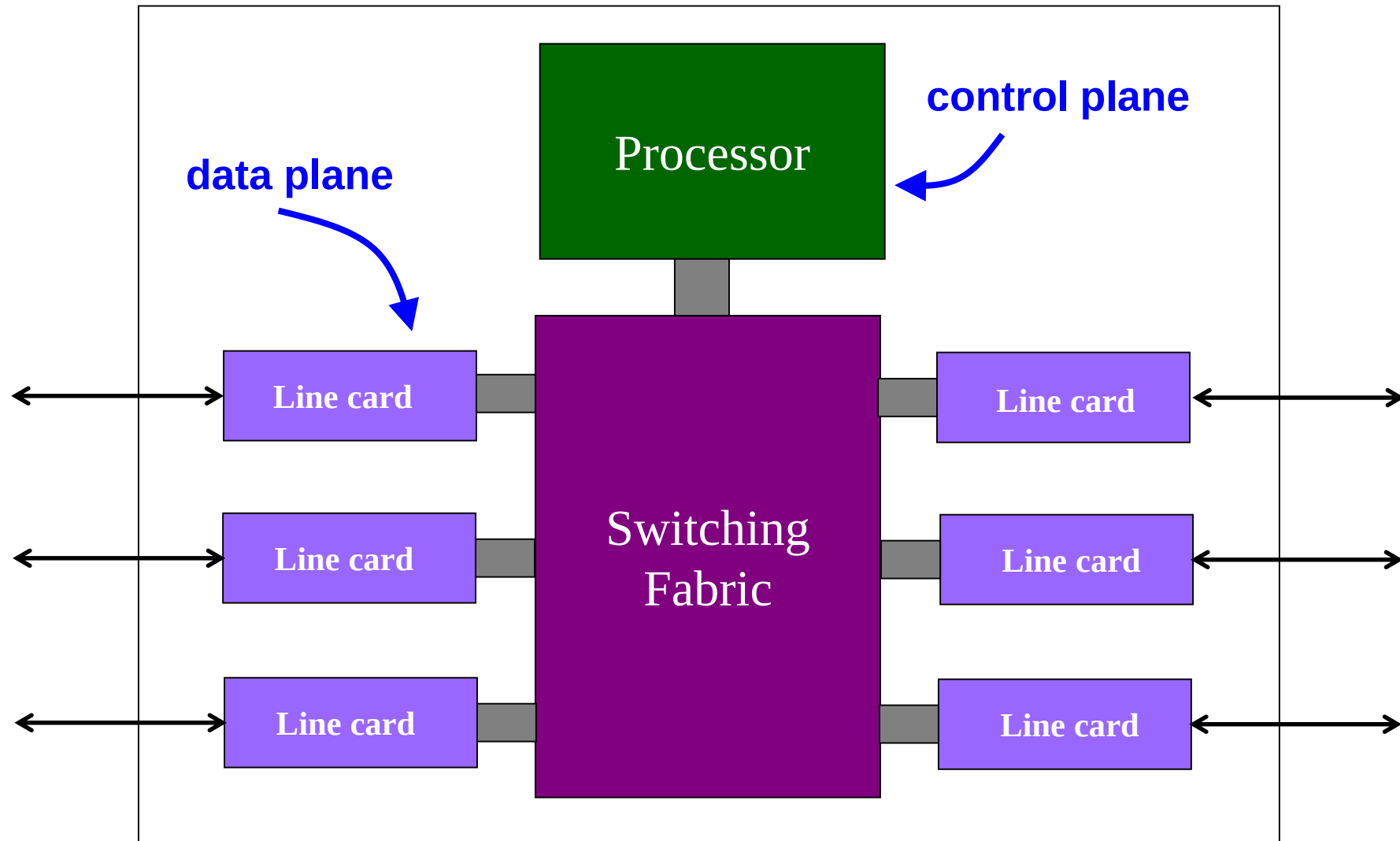
- SDN is an approach to network management that enables dynamic, programmatically efficient network configuration to improve network performance and monitoring.
- In traditional networks, the hardware (like routers and switches) decides how data moves through the network, but SDN changes this by moving the decision-making to a central software system.
- This is done by separating the control plane (which decides where traffic is sent) from the data plane (which moves packets to the selected destination).

Software Defined Networks (SDN)



- 1. Data plane:** processing and delivery of packets with local forwarding state
 1. Forwarding state + packet header → forwarding decision
 2. Filtering, buffering, scheduling
- 2. Control plane:** computing the forwarding state in routers
 1. Determines how and where packets are forwarded
 2. Routing, traffic engineering, failure detection/recovery
- 3. Management plane:** configuring and tuning the network
 1. Traffic engineering, ACL config, device provisioning,

SDN: Data and Control Planes



SDN: Data Planes

- **Data Plane Functions:** The data plane handles user traffic and performs the physical movement of packets through network devices like routers. Its primary tasks include:
 - **Matching on some header bits:** A router examines the destination IP address in the packet's header.
 - **Perform some actions:** Based on the match, the router decides the next step, such as sending the packet to an outgoing link or dropping it.
- **IP Forwarding:** This is the process where a router uses a forwarding table to relay IP packets from one network to another, enabling communication between different network segments, such as a local area network (LAN) and a wide area network (WAN).
- **Forwarding Table:** This table is like a map that stores information about network addresses and their corresponding outgoing interfaces (next hops). Routers use a "longest prefix match" algorithm to find the most specific path for a destination IP address in the table.
- **LAN (Local Area Network):** Connects a small group of devices in a limited geographical area (e.g., in an office or home).
- **WAN (Wide Area Network):** Covers large geographical areas, connecting multiple LANs, often over the internet.

SDN: Data Planes

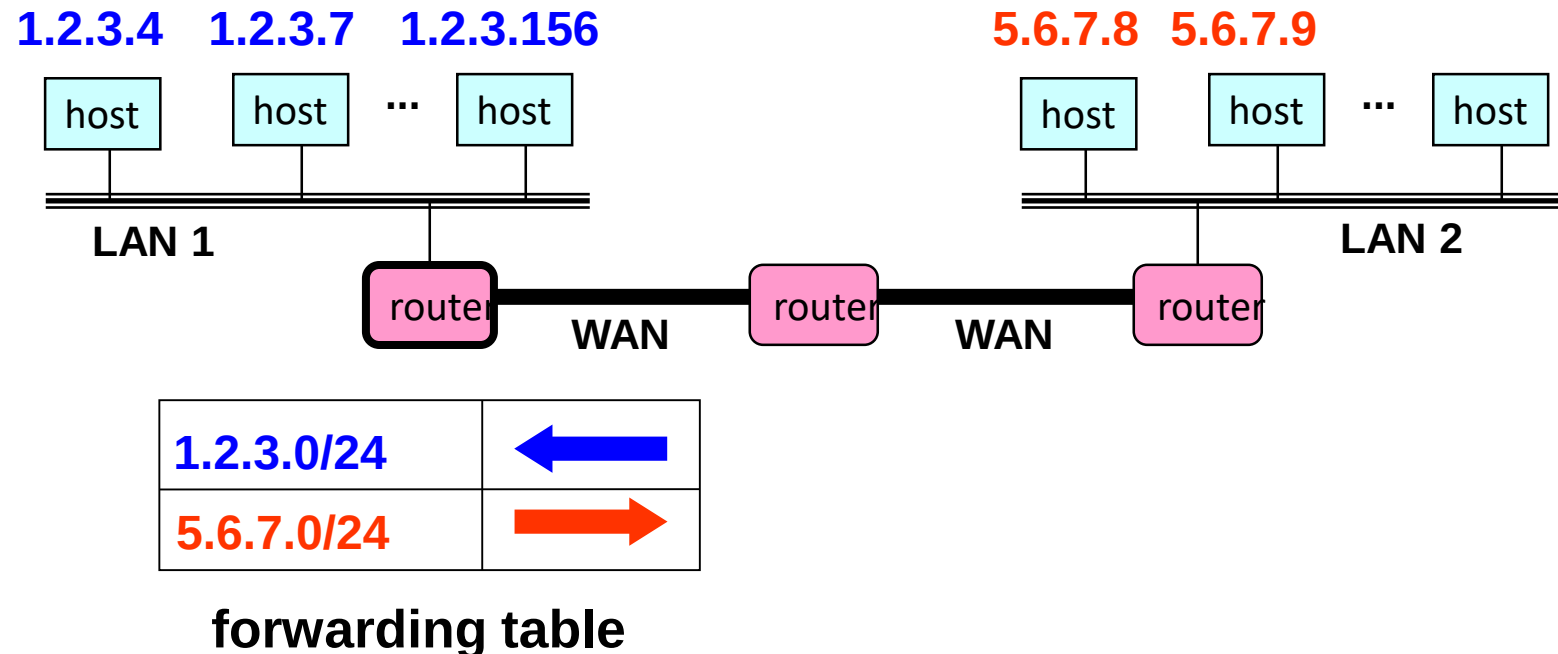
- **Example**

Packet Arrival: A packet arrives at a router's interface, destined for a host in the 5.6.7.0/24 network.

Table Lookup: The router checks its forwarding table, which is built by the control plane. The table has an entry: 5.6.7.0/24 with an orange arrow indicating the outgoing interface.

Forwarding: The router forwards the packet to the correct outgoing interface (via the WAN in this diagram) toward its destination, based on the rules in the table.

The image visually represents this process, showing data moving between two local networks (LAN 1 and LAN 2) interconnected by routers and a wider area network (WAN), guided by the rules in the forwarding table.



SDN: Control Planes

- Compute paths the packets will follow
 - Populate forwarding tables
 - Traditionally, a distributed protocol
- Example: **Link-state routing**
 - Flood the entire topology to all nodes
 - Each node computes shortest paths
 - Dijkstra's algorithm

SDN: Management Plane

- The management plane is where administrators configure and monitor network devices (like routers and switches).
- **Traffic Engineering:** involves influencing the paths data packets take through the network. This is often achieved by assigning numerical "weights" or "metrics" to network links, which are then used by routing protocols to calculate the shortest, or most optimal, path to a destination.
- **Weight Determination:**
 - **Inversely proportional to link capacity:** This is a common heuristic, where faster links (higher capacity) are given lower weights (lower cost) to encourage traffic to use them, which helps to avoid congestion.
 - **Proportional to propagation delay:** Assigning weights based on physical distance or delay is another strategy to ensure traffic takes paths with lower latency.
 - **Network-wide optimization based on traffic:** The most sophisticated approach involves using software tools to model the network's current traffic load and calculate the optimal set of weights to meet specific goals, such as load balancing or guaranteeing a certain quality of service (QoS).

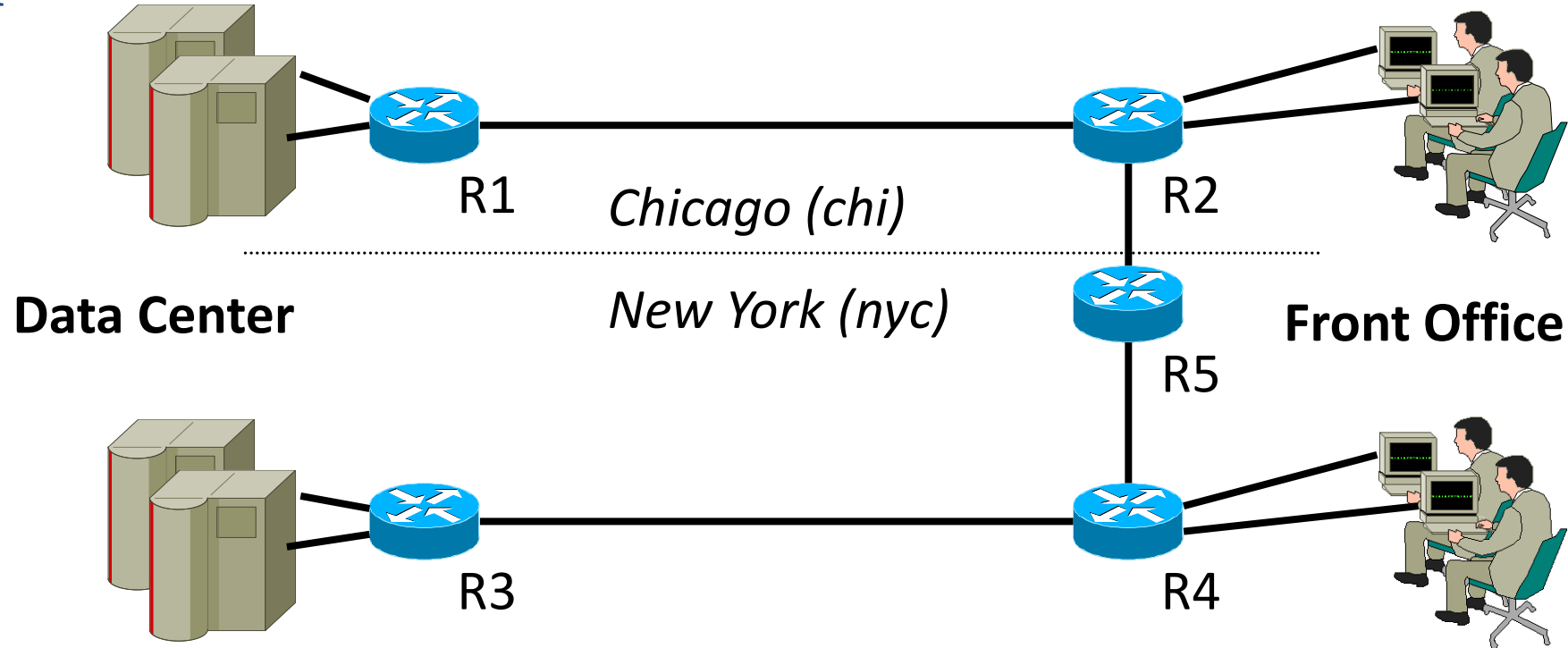
SDN: Challenges

1. (Too) many task-specific control mechanisms
 - No modularity, limited functionality
2. Indirect control
 - Must invert protocol behavior, “coax” it to do what you want
 - Ex. Changing weights instead of paths
3. Uncoordinated control
 - Cannot control which router updates first
4. Interacting protocols and mechanisms
 - Routing, addressing, access control, QoS

The network is

- Hard to reason about
- Hard to evolve
- Expensive

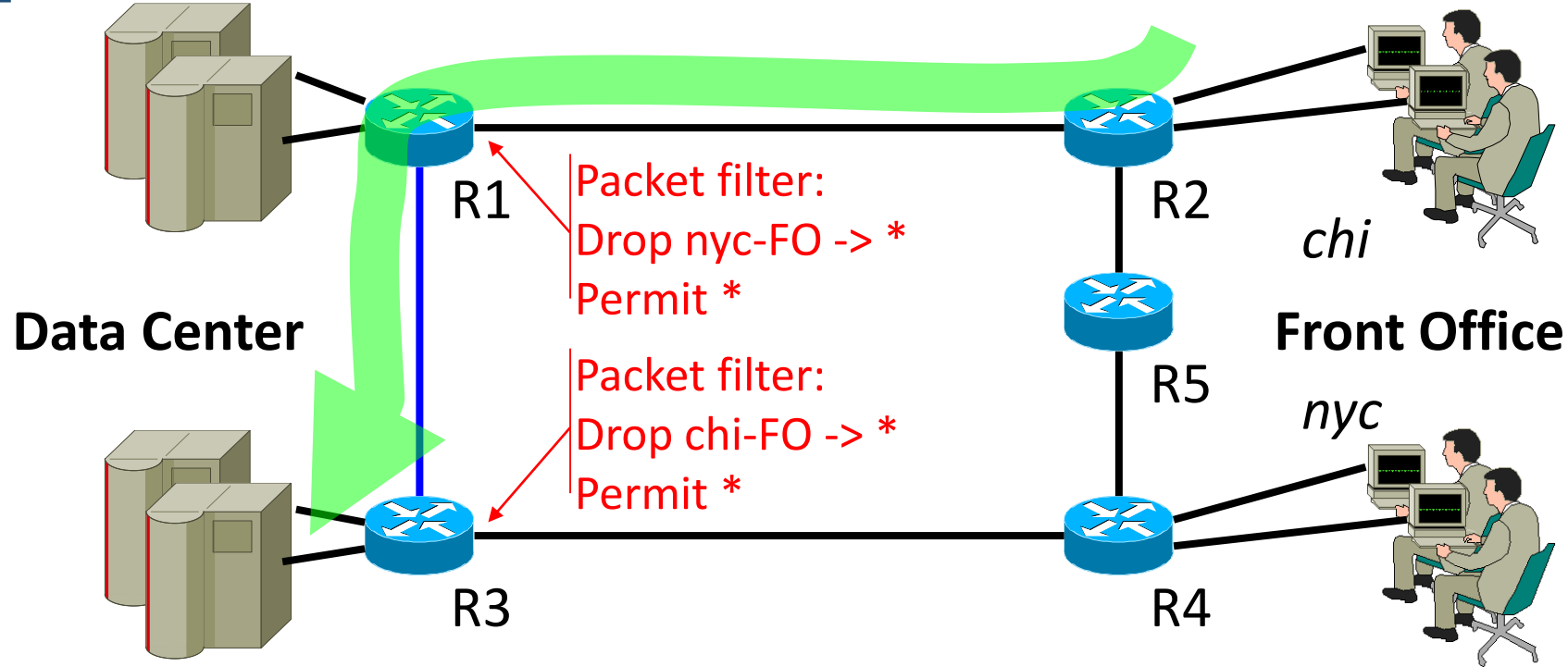
Example: Access Control



- Two locations, each with data center & front office
- All routers exchange routes over all links

	chi-DC	chi-FO	nyc-DC	nyc-FO
chi-DC		●	●	⊘
chi-FO	●		⊘	●
nyc-DC	●	⊘		●
nyc-FO	⊘	●	●	

Example: Access Control



- A new short-cut link added between data centers
- Intended for backup traffic between centers
- **The Conflict:** When a new physical link is added (the red dotted line between R1 and R3), dynamic routing protocols immediately update their paths to take advantage of it. However, the static packet filters (ACLs) are *not* automatically updated. The new, shorter path bypasses the routers where the crucial packet filters were applied, allowing the previously forbidden traffic to now reach the restricted destination.

Wireless Media Access Issues

- IoT is a network of physical objects or devices. These devices exchange data over the internet and can be controlled and monitored remotely. These devices include sensors, wearable devices etc. that can communicate with each other wirelessly. These devices often share same radio frequency spectrum for data transmission.

Wireless Media Access Issues

- **Challenges:**

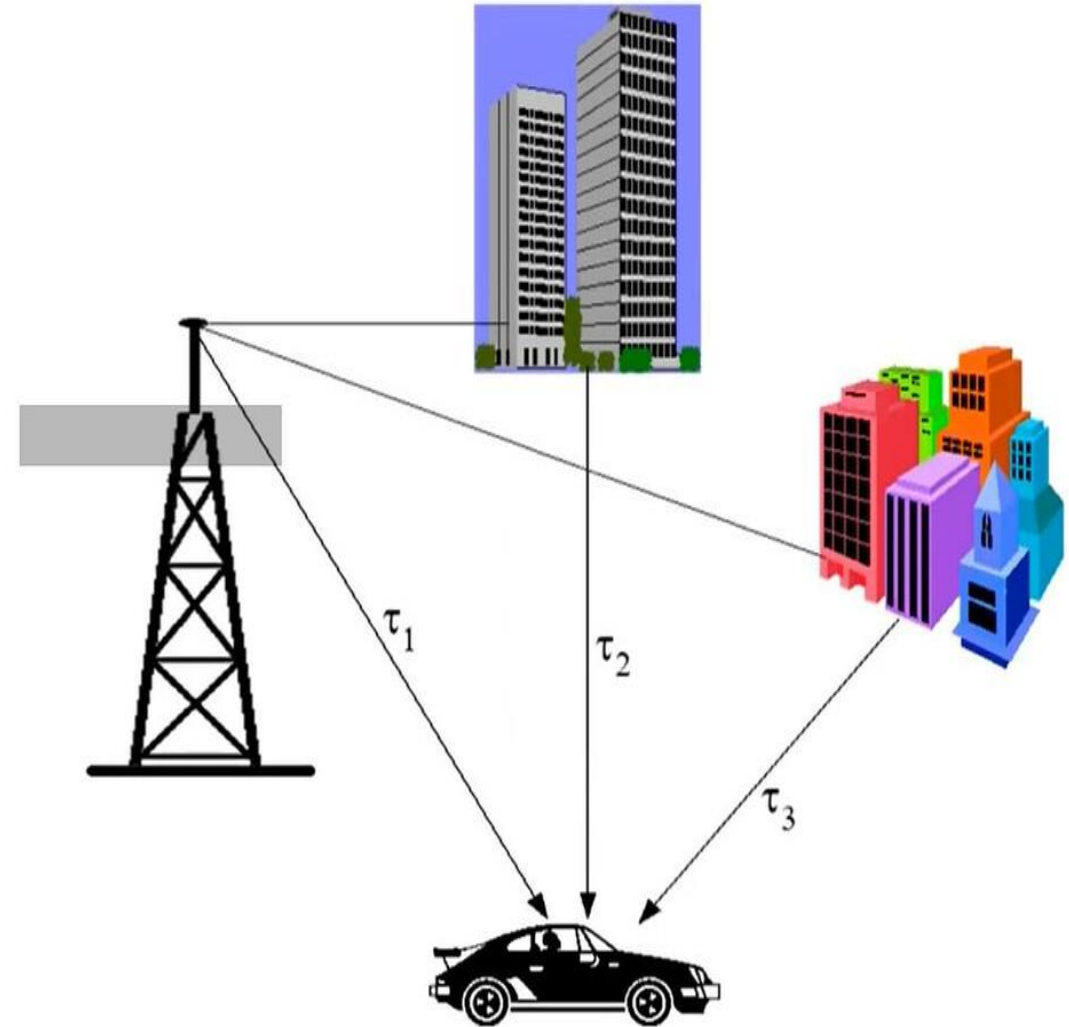
- ✓ **Interference:** Since many IoT devices communicate over the same wireless channels, they can interfere with each other.
- ✓ **Collision Detection:** When two devices try to send data at the same time, their signals can "collide," which causes the data to be lost or corrupted.
- ✓ **Limited Bandwidth:** The amount of wireless space (or bandwidth) available is limited. If too many devices try to communicate at the same time, the network can get congested, slowing down communication.
- ✓ **Energy Consumption:** Many IoT devices, especially battery-powered ones, need to conserve energy. However, communication often requires a lot of power.
- ✓ **Latency:** Some IoT applications, such as real-time monitoring or control systems, require quick communication with very little delay. However, wireless networks may experience delays, or **latency**, which can affect the performance of time-sensitive applications.
- ✓ **Scalability:** IoT networks often have thousands or even millions of devices. Managing the wireless communication between such a large number of devices without creating interference or congestion is challenging.

Wireless Media Access Issues

✓ Time-varying Channel:

- **Reflection** – This occurs when a propagating wave carrying information intrudes on an object that has very large dimensions than the wavelength of the wave.
- **Diffraction** – This occurs when the radio path between the transmitter and the receiver is collided by the surface with sharp edges. This is a phenomenon which causes the diffraction of the wave from the targeted position.
- **Scattering** – This occurs when the medium through from the wave is traveling consists of some objects which have dimensions smaller than the wavelength of the wave.

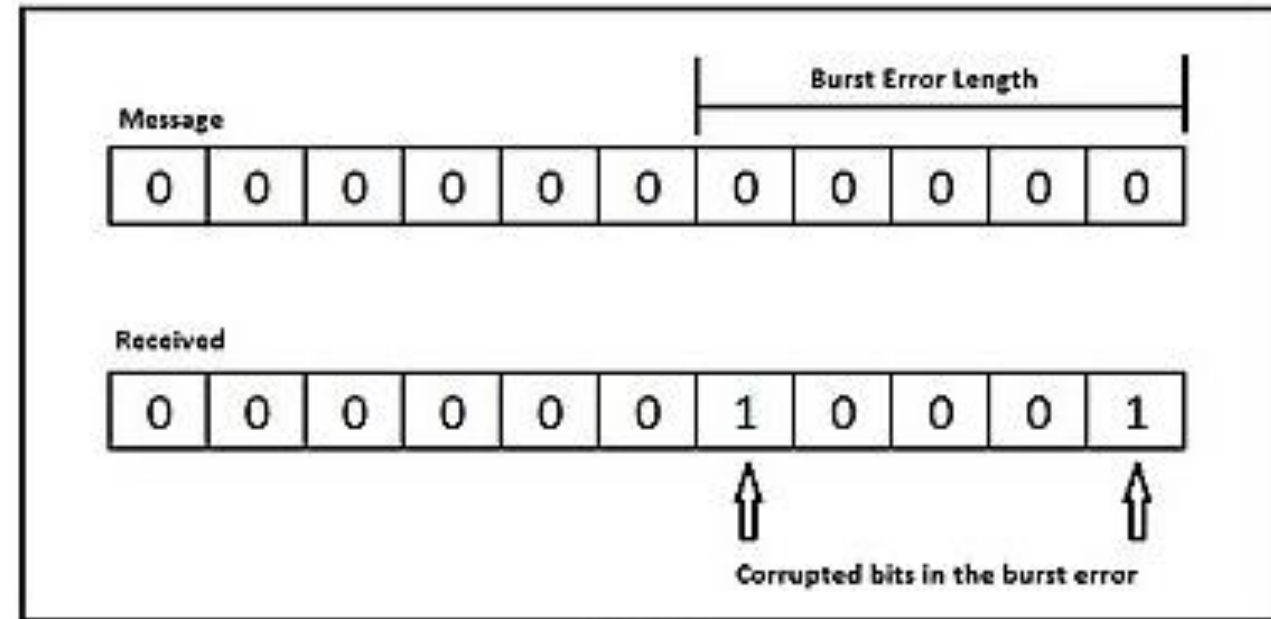
While transmitting the signal by the node these are time shifted and this is called **multipath propagation**. While when this node signals intensity is dropped below a threshold value, then this is termed as **fade**. As a result, **Handshaking strategy** is widely used so as a healthy communication can be set up.



Wireless Media Access Issues

✓ Burst Channel Error:

- Burst channel errors refer to a specific type of error that can occur in communication channels, particularly in digital communication systems such as wireless networks, fiber optic networks, or satellite communication systems.
- In burst channel errors, multiple consecutive bits or symbols are corrupted or lost within a short span of time. This burst of errors can be caused by various factors such as electromagnetic interference, noise, fading, or transmission impairments in the communication medium.
- These errors are particularly problematic because they can disrupt the integrity of transmitted data packets or frames, leading to degradation in the quality of communication or even complete loss of data in severe cases.
- To mitigate burst channel errors, various error correction techniques are employed in communication systems, such as forward error correction (FEC) codes, interleaving, and error detection algorithms. These techniques help to detect and correct errors, or at least mitigate their impact, thus improving the reliability and robustness of the communication system in the presence of burst errors.



MAC Protocol Survey

- In Wireless Sensor Networks (WSNs), the Medium Access Control (MAC) protocol is a set of guidelines that dictate how each node should transmit data over the shared wireless medium.
- The primary objective of the MAC protocol is to minimize the occurrence of idle listening, overhearing, and collisions of data packets. By efficiently managing access to the wireless medium, the MAC protocol helps to reduce energy consumption and optimize the use of network resources.
- **MAC Protocol Categories:**
 1. Contention based MAC
 2. Scheduled based MAC
 3. Hybrid MAC

MAC Protocol Survey: Categories

- **Contention-based MAC:**

- ✓ Contention-based MAC protocol is also known as a random access MAC protocol. It allows all nodes to transmit data on the shared medium, but they have to compete with each other to access the medium. One example of contention-based MAC is **CSMA/CA** (Carrier Sense Multiple Access with Collision Avoidance).
- ✓ In CSMA/CA, each node senses the medium before transmitting the data. If the medium is idle, the node can transmit data immediately. However, if the channel is busy the node has to wait for a random time also known as back-off time. This back-off time reduces the chances of collisions.

- **Scheduled-based MAC:**

- ✓ Scheduled-based MAC is also known as a deterministic MAC protocol. Where each node follows a predetermined schedule and transmits the data according to its given time slot. The data collision is completely nullified in scheduled-based MAC. An example of Scheduled based MAC is **TDMA**(Time Division Multiple Access).
- ✓ In TDMA the time is divided into fixed slots and each node is allocated a specific time frame in which they can transmit the data. During this time slot, other nodes remain silent.

MAC Protocol Survey: Categories

- **Hybrid MAC:**

- ✓ Hybrid MAC is a combination of different protocols such as contention-based MAC and scheduled-based MAC to optimize the performance of wireless sensor networks.
- ✓ For example, contention-based MAC protocols, such as CSMA/CA, allow nodes to access the medium based on a random backoff interval, which reduces collisions but may result in inefficient utilization of the medium. On the other hand, scheduled-based MAC protocols, such as TDMA, divide the medium into time slots and assign them to different nodes, which can achieve high utilization but may not be flexible enough to adapt to changing network conditions.
- ✓ Hybrid MAC solved the issue by using other MAC protocols, During transmission of data if the channel is idle or the channel has low traffic then Hybrid MAC switches to contention-based MAC. If the traffic in the channel increases then it is switched to scheduled-based MAC such as TDMA.

MAC Protocol Survey: Key Considerations

- ✓ **Energy Efficiency:** Critical for battery-powered devices; protocols minimize idle listening, overhearing, and collisions.
- ✓ **Throughput & Delay:** Maximizing data rate while minimizing latency, especially in real-time systems.
- ✓ **Collision Avoidance:** Managing hidden/exposed terminal problems, often using techniques like RTS/CTS handshakes (Request-to-Send/Clear-to-Send).
- ✓ **Scalability & Density:** Performance in dense networks where contention is high.

MAC Protocol in IoT

✓ ALOHA:

- contention-based protocol.
- each node or station transmits a frame without trying to detect whether the transmission channel is idle or busy. If the channel is idle, then the frames will be successfully transmitted. If two frames attempt to occupy the channel simultaneously, collision of frames will occur and the frames will be discarded. These stations may choose to retransmit the corrupted frames repeatedly until successful transmission occurs.
- Pure and Slotted ALOHA: In pure ALOHA, the time of transmission is continuous. Whenever a station has an available frame, it sends the frame. If there is collision and the frame is destroyed, the sender waits for a random amount of time before retransmitting it. Slotted ALOHA reduces the number of collisions and doubles the capacity of pure ALOHA. The shared channel is divided into a number of discrete time intervals called slots. A station can transmit only at the beginning of each slot. However, there can still be collisions if more than one station tries to transmit at the beginning of the same time slot.

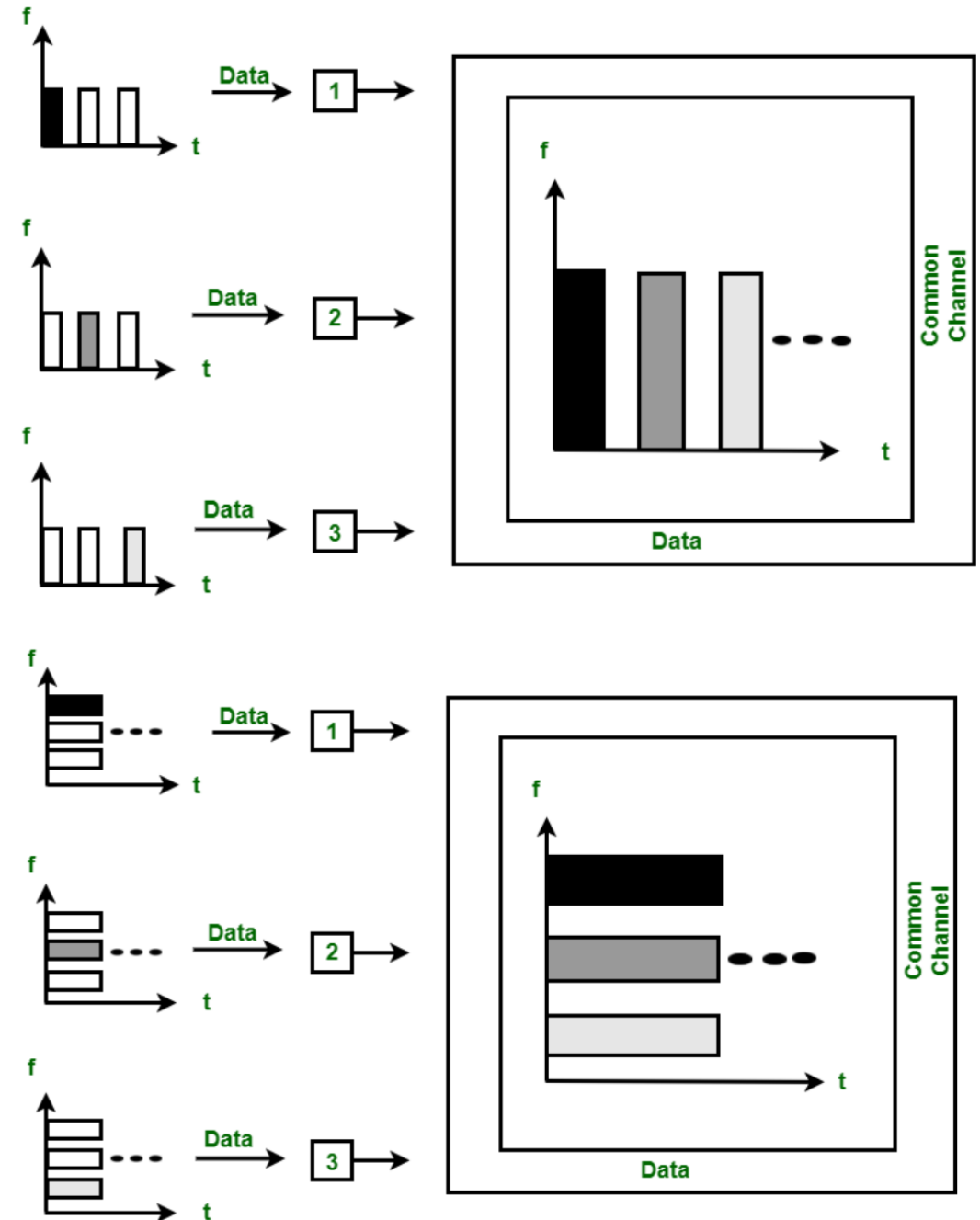
MAC Protocol in IoT

✓ CSMA/CA:

- CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) is a MAC protocol crucial for wireless networks (like Wi-Fi) that prevents data collisions by having devices listen before transmitting and use random backoff timers, rather than detecting collisions after they happen.
- Carrier Sense (CS): A device listens to the wireless channel (senses the carrier frequency) to check if it's busy.
- If Busy: The device waits for a random backoff period before checking again.
- If Idle: The device proceeds to transmit its data packet.
- Collision Avoidance (CA): The random backoff ensures that if multiple devices sense the channel as free, they don't all start transmitting exactly at the same instant, minimizing collisions.
- Acknowledgement (ACK): After transmission, the sender waits for an ACK from the receiver; if none arrives, it assumes a collision or error and retransmits, often with a longer backoff.

MAC Protocol in IoT

- ✓ TDMA/FDMA: Time/Frequency Division Multiple Access; scheduled, collision-free.
 - The TDMA is a protocol in which the bandwidth of channel is divided into the various stations on time basis. There is time slot given to each station, station can transmit data during that time slot only which is as follows. Each station must aware of its beginning of time slot and location of the time slot. TDMA requires synchronization between different stations. It is type of access method in data link layer. At each station data link layer tells station to use allocated time slot.
 - The FDMA is a protocol in which the bandwidth is divided into the various frequency bands. Each station is allocated with band to send data and that band is reserved for particular station for all the time which is as follows. The frequency bands of different stations are separated by small band of unused frequency and that unused frequency bands are called as guard bands that prevent interference of stations.

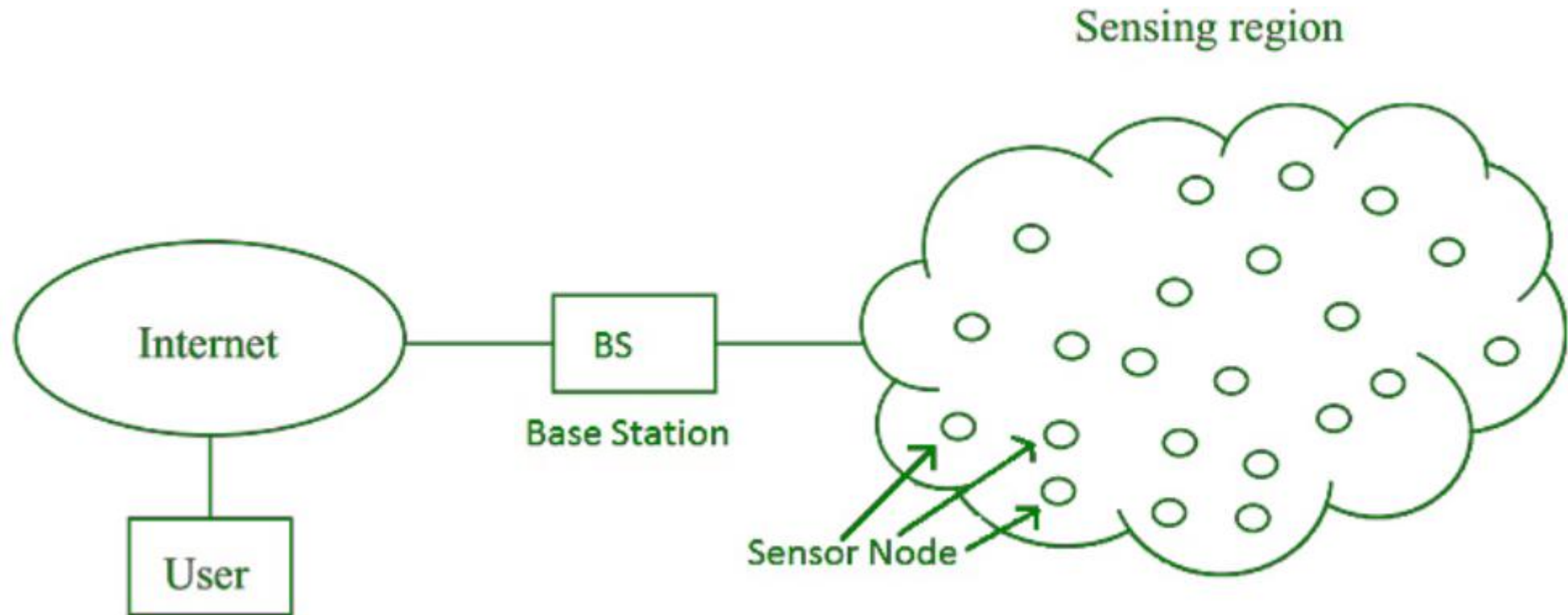


MAC Protocol in IoT

- ✓ S-MAC (Sensor-MAC):
 - Energy-efficient for WSNs, using periodic sleep/wake cycles.
 - It is a contention-based MAC protocol that is specifically designed for wireless sensor networks. The primary objective of SMAC is to minimize idle listening, over-hearing, and collisions of data packets. To achieve this goal, SMAC adopts a duty-cycle approach, also known as a sleep-wakeup cycle. In this approach, each node alternates between a fixed length of active and sleeping periods based on its schedule.
 - To prevent collisions among packets, SMAC utilizes the Request to Send (RTS) and Clear to Send (CTS) packets before transmitting data packets. This helps to ensure that only one node is transmitting data at a time, reducing the likelihood of collisions and improving overall network efficiency.
- ✓ Event-Driven MACs (e.g., SIFT): Tailored for sensor networks, focusing contention on relevant sensors after an event.

Sensor Deployment & Node Discovery

- Sensor nodes are used in WSN with the onboard processor that manages and monitors the environment in a particular area. They are connected to the Base Station which acts as a processing unit in the WSN System. The base Station in a WSN System is connected through the Internet to share data. WSN can be used for processing, analysis, storage, and mining of the data.



Sensor Deployment & Node Discovery

- **Sensor Deployment**

- ✓ The physical placement of sensor nodes in an area of interest to collect data.
- ✓ Types:
 - Deterministic: Fixed, pre-planned positions in accessible areas for optimal configuration.
 - Stochastic/Random: Nodes dropped in hostile environments (e.g., from aircraft).
- ✓ Objectives:
 - Coverage: Ensuring all critical points (targets) are monitored, often using simple, k-coverage (multiple sensors per target), or q-coverage (variable requirements).
 - Connectivity: Ensuring nodes can communicate, often achieved by ensuring sufficient node density or coverage radius.
 - Energy Efficiency: Strategic placement to balance workload and prolong network life.

Sensor Deployment & Node Discovery

- **Node Discovery**

- ✓ The process where nodes find and identify each other and available services after deployment.
- ✓ Key Steps/Concepts:
 - Localization: Nodes determine their own positions (e.g., using Angle of Arrival - AOA) after initial placement.
 - Neighbor Discovery Protocols: Algorithms (like PPR, DPR) used to find nearby nodes, often incorporating energy management to reduce collisions and save power.
 - Service Discovery: Identifying specific functions or services offered by other nodes (e.g., via Bluetooth piconets).
 - Energy Management: Techniques like proxy services on master nodes to let slave nodes sleep, reducing energy drain.
- ✓ Challenges: Communication collisions, ensuring discovery in large networks, and managing limited energy.

Data Aggregation and Dissemination

- **Data Aggregation**

- ✓ **Spatial Aggregation:** Involves combining data from nearby sensors or devices to reduce redundancy and conserve network bandwidth. For example, in environmental monitoring, multiple sensors measuring temperature in close proximity can aggregate their readings to provide an average temperature for the area.
- ✓ **Temporal Aggregation:** Involves aggregating data over time to reduce the frequency of transmissions and save energy. Instead of transmitting data continuously, devices can aggregate data over a certain period (e.g., every minute or hour) and send aggregated summaries.
- ✓ **Statistical Aggregation:** Involves computing statistical measures such as averages, maximums, or minimums from multiple data points. This reduces the volume of data transmitted while preserving important information about the environment.
- ✓ **Hierarchical Aggregation:** Involves aggregating data at different levels of a hierarchy, such as within individual sensors, within local clusters of sensors, and at a central server. This hierarchical approach can balance computational load and communication overhead.

Data Aggregation and Dissemination

- **Data Dissemination:**

- Data dissemination is a process by which data and queries for data are routed in the sensor network.
- In a scope of data dissemination, a source is the node that generates the data and an event is the information to be reported. A node that is interested in data is called sink and the interest is a descriptor for some event that node is interested in.
- After source receives an interest from the sink, the event is transferred from the source to the sink. As a result, data dissemination is a two-step process.
 - ✓ First, the node that is interested in some events, like temperature or air humidity, broadcasts its interests to its neighbors periodically. Interests are then propagated through the whole sensor network.
 - ✓ In the second step, nodes that have requested data send back data after receiving the request. Intermediate nodes in the sensor network also keep a cache of received interests and data.

Data Aggregation and Dissemination

- **Data Dissemination:**

- **Methods:**

- ✓ **Flooding:** In this method, each sensor node that receives a packet broadcasts it to its neighbors assuming that node itself is not the destination of the packet and the maximum hop count is not reached. This ensures that the data and queries for data are sent all over the network
 - In flooding duplicate messages can be sent to the same node which is called implosion. This occurs when a node receives the same message for several neighbors.
 - In addition, the same event may be sensed by several nodes, and thus when using flooding, neighbors will receive duplicate reports of the same event, this situation is called overlap.
 - Many redundant transmissions occur when using flooding and flooding does not take into account available energy at sensor nodes. This wastes a lot of network's resources and decreases the lifetime of the network significantly.

Data Aggregation and Dissemination

- **Data Dissemination:**

- **Methods:**

- ✓ **Gossiping:** Gossiping method is based on flooding, but node that receives the packet forwards it only to a single randomly selected neighbor instead of sending it to all neighbors.

- The advantage of gossiping is that it avoids the problem of implosion and it does not waste as much network resources as flooding.

- The biggest disadvantage of gossiping is that since the neighbor is selected randomly, some nodes in the large network may not receive the message at all. Thus, gossiping is not a reliable method for data dissemination.

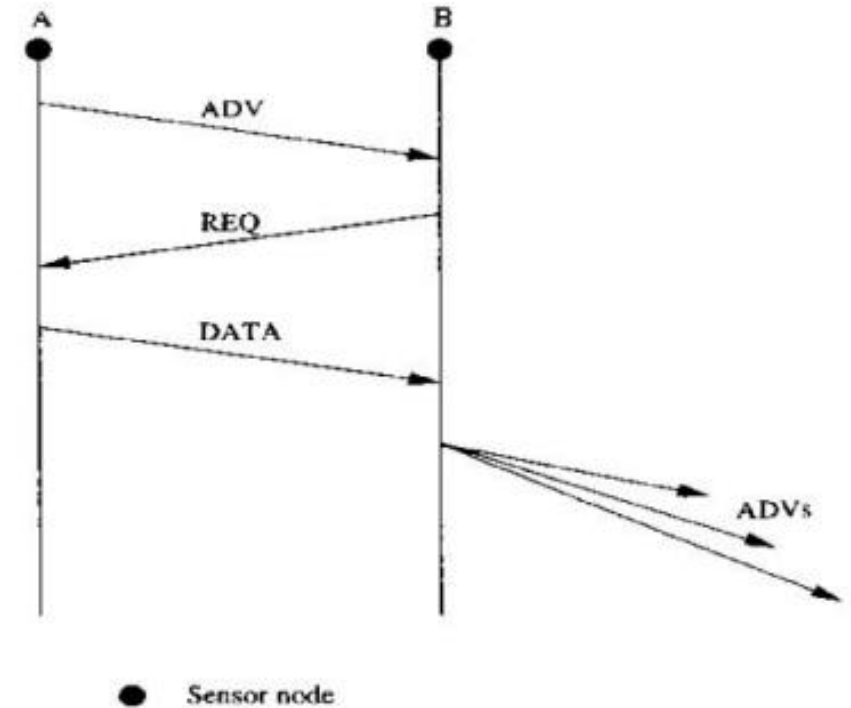
Data Aggregation and Dissemination

- **Data Dissemination:**

- Methods:

- ✓ **SPIN:** Sensor Protocols for Information via Negotiation (SPIN) use negotiation and resource adaption to address the disadvantages of basic flooding.

- SPIN uses data-centric routing, nodes are advertising their data and they will send the data after receiving a reply from interested nodes.
 - SPIN uses three types of messages: ADV, REQ, and DATA.
 - The sensor node that has collected some data sends an ADV message containing meta-data describing the actual data.
 - If some of node's neighbors is interested in the data, the neighbor sends a REQ message back.
 - After receiving the REQ message, the sensor node sends the actual DATA.
 - The neighbor also sends ADV message forward to its neighbors, thus data is disseminated through the network.



Data Aggregation and Dissemination

- **Data Dissemination:**
 - Features:
 - ✓ **Energy Efficiency:** WSNs are typically powered by batteries or other low-power sources, and so energy efficiency is a crucial consideration in data dissemination. Techniques such as data aggregation, compression, and clustering can be used to reduce the amount of data transmitted and conserve energy.
 - ✓ **Reliability:** All nodes in the network receive the necessary data without loss or duplication. Techniques such as error detection and correction can be used to ensure reliability.
 - ✓ **Scalability:** WSNs can range in size from a few nodes to thousands or even millions of nodes. Data dissemination techniques must be scalable to accommodate the number of nodes in the network.
 - ✓ **Security:** WSNs are vulnerable to security threats, such as eavesdropping, tampering, and denial of service attacks. Techniques such as encryption and authentication can be used to ensure the security of data dissemination.
 - ✓ **Adaptability:** WSNs must be adaptable to changing conditions in the environment, such as changes in the number and location of nodes, changes in data rates, and changes in network topology. Techniques such as dynamic routing and load balancing can be used to adapt to these changes.
 - ✓ **QoS Requirements:** In some applications, data dissemination may have Quality of Service (QoS) requirements such as delay, throughput, or reliability. Techniques such as priority scheduling and traffic shaping can be used to ensure that QoS requirements are met.

THANK YOU